Course Logistics, Intro to ML

Tuesday, January 20, 2026 6:01 PM

Agenda
→ Course logistics
→ Intro to ML

Resume @ 6.20 PM
Sreeram.

Adv. GenAI
→ 6 weeks.

Second part - Internship
→ Gen AI

Mod 1) MLops ✓ 1 week

3 weeks ← Mod 2) Building Gen AI ✓ apps. ← Langchain
Gemini
openAI
→ Hugging face (Models)
RAG

Mod 3) Agentic AI ✓
→ 2 weeks.

Mod 1: MLOps:
→ What is ML, Why ML

ML Model building pipe line ≈ 9 steps
+ optimizing → Hand on
each step
→ Diff operations
→ How to implement

→ Hyper param Tuning
→ (a lot of data)
a lot of expt
→ Manually - Difficult to manage

ML - LR → 10
Log R → 10
KNN → 10
DT → 10

→ Expt Tracking ⇒ using ML Flow → DL → Tensor board
wandb.ai
→ DL, GenAI
→ good number of Models
→ ✓ Model Management using ML flow
→

Deploy Model
Frontend | UI
→ Deploy Model

Deploy Model

Prefect → Manage Multiple stages.

Grokking Machine Learning by Luis Serrano.

Module 2:

Building GenAI applications

What is GenAI, LLM.

API → OpenAI API ← GPT Models

Gemini API → use Gemini Models ✓ Freeher

Build simple chatbot applcn

Try to understand

→ How generation works?

Different param (Text-gen)

Temperature

Top-p

Top-k

Prompt Engineering

→ Good / Better performing applcn

Any applcn

GenAI

Agent AI } i/p → prompt

Must know.

→ Good quality prompt as i/p

→ good quality output

Lang Chain

→ promp Templates

Model ← api → replace → Models

Output parser

→ Hugging face

Add External tools, Memory

→ To applcn built

Retrival Augmented ← RAG applications

Retrieval Augmented Generation ← RAG applications

Mod 3: Agentic AI — 2 weeks

5, 6 ↙ ↘

Lang graph

Crew AI

Autogen

typing module

Build a graph

↓

Simple Agent

+ Tools

+ Memory < Short / LT / external mem (SQL)

Types of o/p gen < invoke (@ end) / Stream (ChatGPT)

Human in the loop (HITL) { Human Intervent / Dec ← Breakpoints

↓

Interactions b/w Multiple Agents

For every weeks → As an ent: Class dis → Give Homeworks atleast two

@ end every topic → project

Machine Learning → predictions.

↓ what?

↓ Making M/c Learn Data

↓ Learn the patterns

Machine → Even with ML → Draw patterns → Actionable Insights

Data Anal → Draw pattern → Actionable Insights → why ML?

why? → Handle Complex

Data Analy → Desc / Int → Limited to Simple Datasets → Linear / Simple

Actionable Insights

Learn the patterns from the Data

Input → Models → Output | Insights

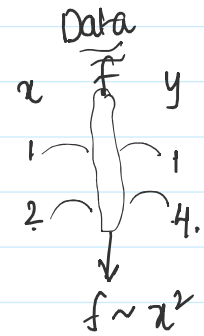
Model ≈ Magic | Complex

$y = f(x)$

$y = x^2$ → Model

Handle Complex Datasets → Draw patterns from Complex → ML needed

Linear/Simple patterns from data



Model → Math representation of reality.

Study a phenomenon

How Gen AI helps Human beings? → Data → Table

Learn patterns → Model → $y = f(x)$. Model

Model Building pipeline

1) Load Data

2) EDA ⇒ Validity data

3) Segregate features & Target

4) Divide Data into Train & Test

5) Data (Train) pre processing

6) Build a model

7) Data (Test) pre processing

8) Evaluation of Model

9) Deployment